Flexible demand: solutions

These solutions are written to be read after the workshop and follow the parts of the worksheet. Equation numbers (11)–(17) are those of the lecture slides, which follow Ambec and Crampes (2021).

Solutions 1 · The planner

Group C, Q1: three conditions, one for each consumption choice

Each consumption variable appears in gross surplus and in market clearing, q_r^h once in each and $q_{\overline{r}}$ in both market-clearing constraints. The derivatives are

$$\frac{\partial \mathcal{L}}{\partial q_r^w} = \beta\nu S'(q_r^w) - \beta\nu\gamma_w = 0$$
$$\frac{\partial \mathcal{L}}{\partial q_r^{\overline{w}}} = \beta(1-\nu)S'(q_r^{\overline{w}}) - \beta(1-\nu)\gamma_{\overline{w}} = 0$$
$$\frac{\partial \mathcal{L}}{\partial q_{\overline{r}}} = (1-\beta)S'(q_{\overline{r}}) - \nu\gamma_w(1-\beta) - (1-\nu)\gamma_{\overline{w}}(1-\beta) = 0$$

Dividing out the weights $\beta\nu$, $\beta(1-\nu)$ and $1-\beta$, which are positive because $0 < \beta < 1$ and $0 < \nu < 1$, gives

$$S'(q_r^w) = \gamma_w \tag{12}$$
$$S'(q_r^{\overline{w}}) = \gamma_{\overline{w}} \tag{13}$$
$$S'(q_{\overline{r}}) = \nu\gamma_w + (1-\nu)\gamma_{\overline{w}} \tag{14}$$

Group P, Q1: two conditions for thermal output and capacity

q_f^w appears in the fuel cost in w, in market clearing in w, and in the two bounds $\underline{\mu}_f q_f^w$ and $\overline{\mu}_f(K_f - q_f^w)$. K_f appears in the fuel cost in $\overline{w}$ (where thermal runs at full capacity), in the capacity cost $r_f K_f$, in market clearing in $\overline{w}$, and in the upper bound. The derivatives are

$$\frac{\partial \mathcal{L}}{\partial q_f^w} = -\nu(c+\delta) + \nu\gamma_w + \nu\underline{\mu}_f - \nu\overline{\mu}_f = 0$$
$$\frac{\partial \mathcal{L}}{\partial K_f} = -(1-\nu)(c+\delta) - r_f + (1-\nu)\gamma_{\overline{w}} + \nu\overline{\mu}_f = 0$$

Dividing the first by ν and collecting terms in the second gives

$$-(c+\delta) + \gamma_w - \overline{\mu}_f + \underline{\mu}_f = 0 \tag{15}$$
$$(1-\nu)(\gamma_{\overline{w}} - c - \delta) - r_f + \nu\overline{\mu}_f = 0 \tag{16}$$

Group C, Q2: reactive consumers face the state price, the others the expected price

In each condition, marginal utility equals the marginal social cost of electricity. A reactive consumer chooses one quantity for each state, so by (12) and (13) it should face the state price γ_h. The two conditions are its inverse demand in each state. A non-reactive consumer chooses one quantity for both states. One more unit costs γ_w with probability ν and $\gamma_{\overline{w}}$ with probability $1-\nu$, so the price in (14) is the expected price. It equals the fair flat retail price $p = \nu p^w + (1-\nu)p^{\overline{w}}$.

Group P, Q2: run thermal when the price covers its cost, and build it until a MW pays for itself

(15) is the merit order. Thermal output is used when the price covers fuel and carbon, $\gamma_w \geq c + \delta$. If the price is higher, thermal runs at full capacity and earns the scarcity rent $\overline{\mu}_f = \gamma_w - c - \delta$, its margin above

cost. If it is lower, thermal does not run ($\underline{\mu}_f > 0$). (16) says that the planner adds thermal capacity until the expected net earnings of one more MW equal its cost r_f. The MW earns $\gamma_{\overline{w}} - c - \delta$ in the state without wind and the scarcity rent $\overline{\mu}_f$ in the windy state.

Which state pays for a MW of thermal capacity? Suppose capacity is not binding in the windy state, $\overline{\mu}_f = 0$. Then (16) gives

$$\gamma_{\overline{w}} = c + \delta + \frac{r_f}{1-\nu}.$$

A MW of thermal capacity earns a margin above $c + \delta$ only in the state without wind. That state occurs with probability $1 - \nu$, so the margin there must be $r_f/(1-\nu)$ for the expected margin to equal r_f. This is peak-load pricing (Steiner 1957, Boiteux 1960): the state that needs the capacity pays for it, and the rarer that state is, the higher the price when it occurs.

This assumes that thermal capacity is idle in part of the windy state (cases b2 and c of Proposition 1). When it runs at full capacity in both states (cases a and b1), $\overline{\mu}_f = \gamma_w - c - \delta > 0$ and the windy state pays part of the capacity too: (16) says that each state pays with its margin above $c + \delta$.

If you finish early: the last turbine just pays for itself

What $\underline{\mu}_i$ is. The planner builds every wind site with capacity cost below the threshold $\tilde{r}_i$. The threshold cannot go below the cheapest site, $\tilde{r}_i \geq \underline{r}_i$. At $\tilde{r}_i = \underline{r}_i$, $F(\underline{r}_i) = 0$ and no wind is built. The constraint enters the Lagrangian as $\nu\underline{\mu}_i(\tilde{r}_i - \underline{r}_i)$, and $\underline{\mu}_i$ is its shadow price. The Kuhn–Tucker conditions are

$$\underline{\mu}_i \geq 0, \qquad \tilde{r}_i - \underline{r}_i \geq 0, \qquad \underline{\mu}_i(\tilde{r}_i - \underline{r}_i) = 0.$$

The last one says that $\underline{\mu}_i$ can be positive only when the bound binds. Because the constraint has the weight ν in front, the welfare gain from relaxing the bound by one unit is $\nu\underline{\mu}_i$.

Where $\tilde{r}_i$ appears. In three places in $\mathcal{L}$:

$$-\overline{K}\int_{\underline{r}_i}^{\tilde{r}_i} r_i\, dF(r_i), \qquad \nu\gamma_w\overline{K}F(\tilde{r}_i), \qquad \nu\underline{\mu}_i(\tilde{r}_i - \underline{r}_i).$$

The capacity cost: Leibniz' rule. Write $dF(r_i) = f(r_i)\, dr_i$, so the integral is $\int_{\underline{r}_i}^{\tilde{r}_i} r_i f(r_i)\, dr_i$. Leibniz' rule for an integral whose limits depend on x is

$$\frac{d}{dx}\int_{a(x)}^{b(x)} g(r,x)\, dr = g\big(b(x),x\big)\, b'(x) - g\big(a(x),x\big)\, a'(x) + \int_{a(x)}^{b(x)} \frac{\partial g(r,x)}{\partial x}\, dr.$$

Here $x = \tilde{r}_i$, the upper limit is $b = \tilde{r}_i$ with $b' = 1$, the lower limit $\underline{r}_i$ is fixed so $a' = 0$, and the integrand $g = r_i f(r_i)$ does not depend on $\tilde{r}_i$, so the last term is zero. Only the first term is left:

$$\frac{d}{d\tilde{r}_i}\int_{\underline{r}_i}^{\tilde{r}_i} r_i f(r_i)\, dr_i = \tilde{r}_i f(\tilde{r}_i).$$

The cost of the marginal site is $\tilde{r}_i$, and $\overline{K}f(\tilde{r}_i)$ units of wind capacity are added when the threshold rises by one unit. The derivative of the first term is therefore $-\overline{K}\tilde{r}_i f(\tilde{r}_i)$.

The other two terms. The derivative of F is the density f, so market clearing gives $\nu\gamma_w\overline{K}f(\tilde{r}_i)$. The bound gives $\nu\underline{\mu}_i$.

Collect. Adding the three and setting the sum to zero:

$$-\overline{K}\tilde{r}_i f(\tilde{r}_i) + \nu\gamma_w\overline{K}f(\tilde{r}_i) + \nu\underline{\mu}_i = 0 \quad \Leftrightarrow \quad \overline{K}f(\tilde{r}_i)\,(\nu\gamma_w - \tilde{r}_i) + \nu\underline{\mu}_i = 0.$$

Assume that the density is positive, $f(r_i) > 0$, on $[\underline{r}_i, \infty)$. Dividing by $\overline{K}f(\tilde{r}_i) > 0$ gives (17) on the slides, $-\tilde{r}_i + \nu\gamma_w + \nu\underline{\mu}' = 0$ with $\underline{\mu}' = \underline{\mu}_i/\big(\overline{K}f(\tilde{r}_i)\big)$.

Interpretation. One more unit of wind capacity produces only in the windy state, so it earns γ_w with probability ν. The marginal site costs $\tilde{r}_i$. There are two cases.

- Some wind is built ($\tilde{r}_i > \underline{r}_i$). Then $\underline{\mu}_i = 0$ and $\tilde{r}_i = \nu\gamma_w$: the last turbine built costs exactly what it earns in expectation. A cheaper site earns more than it costs, and a more expensive one less.
- No wind is built ($\tilde{r}_i = \underline{r}_i$). Then $\underline{\mu}_i = \overline{K} f(\underline{r}_i)(\underline{r}_i - \nu\gamma_w)/\nu$. This is positive when $\nu\gamma_w < \underline{r}_i$, that is, when even the cheapest site earns less than it costs.

Solutions 2 · A two-state example

Demand. With $S(q) = 200q - 5q^2$, marginal utility is $S'(q) = 200 - 10q$. A consumer buys until marginal utility equals the price, $200 - 10q = p$, so $q = 20 - 0.1p$ and $D'(p) = -0.1$.

Q1: the reactive consumer has less gross surplus

The reactive consumer faces 40 in state w and 120 in state $\overline{w}$. The non-reactive consumer pays 80 in both.

	Price	Quantity $q = 20 - 0.1p$	Gross surplus $S(q)$
Reactive, state w	40	16	$3200 - 1280 = 1920$
Reactive, state $\overline{w}$	120	8	$1600 - 320 = 1280$
Non-reactive, both states	80	12	$2400 - 720 = 1680$

The reactive consumer's expected gross surplus is $\frac{1}{2} \cdot 1920 + \frac{1}{2} \cdot 1280 = 1600$. The non-reactive consumer has 1680, so the reactive consumer has 80 less. The reason is that S is concave and that, with linear demand, the reactive consumer buys 12 on average, as the non-reactive one does, but spread unevenly over the two states. (Lemma 2 states the condition as $S(D(p))$ concave in p. Here $S(D(p)) = 2000 - 0.05p^2$.) Compared with the non-reactive consumer, the reactive one gains 4 units in w, worth $1920 - 1680 = 240$, and loses 4 units in $\overline{w}$, worth $1680 - 1280 = 400$. The units lost are worth more than the units gained, and the expected difference is $\frac{1}{2} \cdot 240 - \frac{1}{2} \cdot 400 = -80$. This is Lemma 2, and -80 is the first line of (11).

Q2: moving consumption to the cheap state saves 160

The second line of (11) measures what it costs to supply the reactive consumer compared with the non-reactive one, with a minus sign in front.

State	Probability	Price	$q_r^h - q_{\overline{r}}$	Expected extra cost
w	$\frac{1}{2}$	40	$16 - 12 = +4$	$\frac{1}{2} \cdot 40 \cdot 4 = 80$
$\overline{w}$	$\frac{1}{2}$	120	$8 - 12 = -4$	$\frac{1}{2} \cdot 120 \cdot (-4) = -240$
Total				-160

The reactive consumer costs 160 less to supply, so the two price terms in (11) sum to $+160$. The sum is positive because the reactive consumer moves consumption from the state where electricity is expensive to the state where it is cheap. Adding the two lines of (11):

$$\frac{d\mathbb{E}[W]}{d\beta} = -80 + 160 = 80.$$

The formula from the lecture gives the same number. The variance of the price is $\mathrm{Var}(p^h) = \frac{1}{2} \cdot \frac{1}{2} \cdot (40 - 120)^2 = 1600$, so

$$\tfrac{1}{2}|D'(p)|\,\mathrm{Var}(p^h) = \tfrac{1}{2} \cdot 0.1 \cdot 1600 = 80.$$

Why the formula works. Look at each state separately. In w the price is 40 below the retail price of 80, and the reactive consumer buys 4 more units. Each extra unit is worth between 80 and 40 to the consumer and costs 40 to supply, so the gain is a triangle with height 40 and width 4, $\frac{1}{2} \cdot 40 \cdot 4 = 80$. In $\overline{w}$ the price is 40 above 80, and the consumer gives up 4 units that are worth between 80 and 120 to it but cost 120 to supply. That is also a triangle, $\frac{1}{2} \cdot 40 \cdot 4 = 80$. The expected gain is $\frac{1}{2} \cdot 80 + \frac{1}{2} \cdot 80 = 80$. Each triangle is the deadweight loss of the fixed price in that state, which RTP removes. The consumer collects it in expectation: the payment rectangles cancel because the retail price is the expected price. With linear demand their sides are straight, which is why the formula is exact here.

Q3: halving the price gap cuts the gain to a quarter

With prices 60 and 100 the reactive consumer buys 14 and 10. The non-reactive consumer still pays 80 and buys 12.

	Quantity	Gross surplus	Expected extra cost
Reactive, state w (price 60)	14	1820	$\frac{1}{2} \cdot 60 \cdot 2 = 60$
Reactive, state $\overline{w}$ (price 100)	10	1500	$\frac{1}{2} \cdot 100 \cdot (-2) = -100$
Non-reactive (price 80)	12	1680	

The first line of (11) is $\frac{1}{2} \cdot 1820 + \frac{1}{2} \cdot 1500 - 1680 = -20$. The reactive consumer costs 40 less to supply, so the second line is +40. Then

$$\frac{d\mathbb{E}[W]}{d\beta} = -20 + 40 = 20 = \tfrac{1}{2} \cdot 0.1 \cdot \tfrac{1}{4} \cdot (60 - 100)^2.$$

Each triangle now has height 20 and width 2, so it is $\frac{1}{2} \cdot 20 \cdot 2 = 20$. Halving the price gap halves both the height and the width, and the area falls to a quarter. This is Proposition 3. As more consumers become reactive, the prices in the two states move closer together, and the next reactive consumer adds less. One consumer is too small to move the prices, so in each calculation they are taken as given. The question assumes that both prices move while the retail price stays at 80. In Proposition 1 that is case b1, where thermal runs at full capacity in both states. In b2 and c, the price in the state without wind is fixed by (16).

If you finish early: a Spanish household gains about €0.29 a year

The slope of demand is $|D'(p)| = |\varepsilon|\, q/p = 0.1 \cdot 0.24/10.82 \approx 0.0022$ kWh per hour per c/kWh, and the variance of the price is $1.73^2 \approx 2.99$. The gain is

$$\tfrac{1}{2} \cdot 0.0022 \cdot 2.99 \approx 0.0033 \text{ cents per hour,}$$

or $0.0033 \cdot 8{,}760 \approx 29$ cents, about €0.29 a year. With an elasticity ten times as large ($\varepsilon = -1$), it is about €2.91 a year. Following the hourly price is not worth the effort for a household. This supports explanation 1 in the lecture: consumers are rationally price inelastic. Fabra et al. (2021, p. 428) read their results the same way. They compute bill savings instead: moving 10% of daily consumption to the cheapest hour saves €0.19 a month, about €2.30 a year. That is a saving on the bill, not a welfare gain, since it ignores the cost of moving the consumption, and even so it is small. Applying the two-state formula across all hours is our own extension. As the worksheet says, the standard deviation is over all hours in the sample, and $\varepsilon = -0.1$ is larger than the paper's own estimates, so treat the result as an upper bound.

Solutions 3 · The grid

Q1: a typical rule is a time window

Typical answers are to charge after midnight, or to avoid 17–21, when the distribution tariff has its peak. Both rules are fixed in advance by the clock.

Q2: if all cars follow the rule, the peak moves

If every car that is plugged in starts when the low tariff begins at 00:00, the peak is 9.33 GW at night. That is higher than the 7.93 GW when the cars charge as people come home, and 2.5 GW above the 6.82 GW peak without the cars. A rule that is grid-friendly for one car stops being grid-friendly when every car follows it, because it moves the peak to the hour the rule points to. The figure is an upper bound, since not all cars start at the same minute, but any fixed window has the same effect on a smaller scale.

Q3: a local, dynamic grid price would make the rule unnecessary

The price that would make the rule unnecessary is the shadow price of the grid, λ_h, at the place where the car is connected and at the time the grid is full. Consumers would then spread their charging by themselves, because the price rises in the hours when the local grid binds. The lecture gives two reasons why this is hard in Denmark. Distribution tariffs use the same time-of-use windows in the whole country, with fixed ratios between them, set in advance from the households' load profile; Energinet's transmission tariff is flat over the day. And the Electricity Supply Act (§ 73, stk. 1) allows geographically differentiated tariffs for consumers only at 10 kV and above and for local associations of network users.

If you finish early: a flat tariff makes reactive consumers use too much when the grid is full

(i) With $\lambda_{\overline{w}} = 0$, the efficient condition for non-reactive consumers is $S'(q_{\overline{r}}) = \nu(\gamma_w + \lambda_w) + (1-\nu)\gamma_{\overline{w}}$. With the tariff they pay $\nu\gamma_w + (1-\nu)\gamma_{\overline{w}} + t$. The two are equal when

$$t = \nu\lambda_w.$$

The flat tariff is the expected grid price, in the same way as the flat retail price is the expected energy price.

(ii) A reactive consumer sees $\gamma_h + t$:

State	Price seen	Efficient price	Difference
w (grid full)	$\gamma_w + \nu\lambda_w$	$\gamma_w + \lambda_w$	$(1-\nu)\lambda_w$ too low
$\overline{w}$ (grid not full)	$\gamma_{\overline{w}} + \nu\lambda_w$	$\gamma_{\overline{w}}$	$\nu\lambda_w$ too high

Reactive consumers use too little when the grid has spare capacity and too much when it is full. The energy part of their price is right, and the grid part is wrong in both states.

Hint B, step 4. A Danish household on a spot-price contract sees γ_h plus a tariff fixed in advance by time of day. That is closer to t than to λ_h. The tariff varies over the day, but it does not depend on whether the grid is actually full, and it is the same everywhere in the grid company's area.